

DANIEL B. JOHNSON

MATERIALS ENGINEER

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SUMMARY

Materials Science major at the University of Michigan. Experience in materials characterization, engineering simulations & computational methods, data processing & visualization, and robotics/electronics.

EDUCATION

Bachelors in Materials Science & Engineering | University of Michigan, Ann Arbor Aug 2023- Dec 2026

- Coursework in polymers, electrical materials, biomaterials, characterization, FEA simulations, etc.
- Characterizing material properties with SEM, OM, DSC, XRD, FTIR, UV-Vis Spec, and range of other analyses.

WORK EXPERIENCE

Materials Science Intern | Bluuzone Bottles Feb 2025 - June 2025

- Codified key material testing processes for mycelium-based disposable water bottles.
- Wrote technical overviews and grant applications to communicate research and secure funding.
- Communicated progress to technical and nontechnical team members to ensure optimal synergy.

Lab Researcher | EMBiR Lab & BIM Lab, University of Michigan Aug 2023 - Present

- Studying viper fangs' material properties to simulate internal venom flow with CFD.
- Testing rheology to determine extent of Newtonian or non-Newtonian behavior in experimental fluids.
- Building CNC indenter to automate non-destructive property measurements like elastic modulus.
- Published findings on novel geometries for building soft biomimetic electrostatic actuators.

Audio Staff | Duderstadt Audio Studios Aug 2024 - Present

- Trained in safe and effective use of analog and digital tools for professional audio recording.
- Identified and resolved conflicts in fast-paced recording sessions to assure client satisfaction.

LEADERSHIP

Operations Lead | Michigan Carbon Capture Aug 2024 - Present

- Designing microbe-powered device capable of wastewater CO₂ capture and H₂ gas production.
- Managing 20-person project team's finances, logistics, recruitment, and yearly growth.
- Considering chemistry of electrolysis cells to maximize yield and minimize harmful byproducts.

Managing Investigative Editor | The Michigan Daily Student Newspaper Feb 2024 - Present

- Leading 10-person team in long-term investigations of wrongdoing by U-M and Ann Arbor institutions.
- Creating data visualizations following best numerical practices with tools like Pandas, D3, and R.
- Writing and editing articles, focusing on clear and concise communication of technical subjects.

TECHNICAL SKILLS

Software: FEA, CFD, CAD (SOLIDWORKS), Python, MATLAB, R, C++, MeshLab, Unity, Blender, HTML/CSS/JS

Hardware: SEM, FTIR, DSC, XRD, Optical microscopy, Tensile/compression testing, Hardness testing, UV-Visible Spectroscopy, Mill/lathe machining, Silicone molding, Microbial culture